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Muscle memory in humans: evidence for myonuclear permanence and long-term transcriptional regulation after strength training

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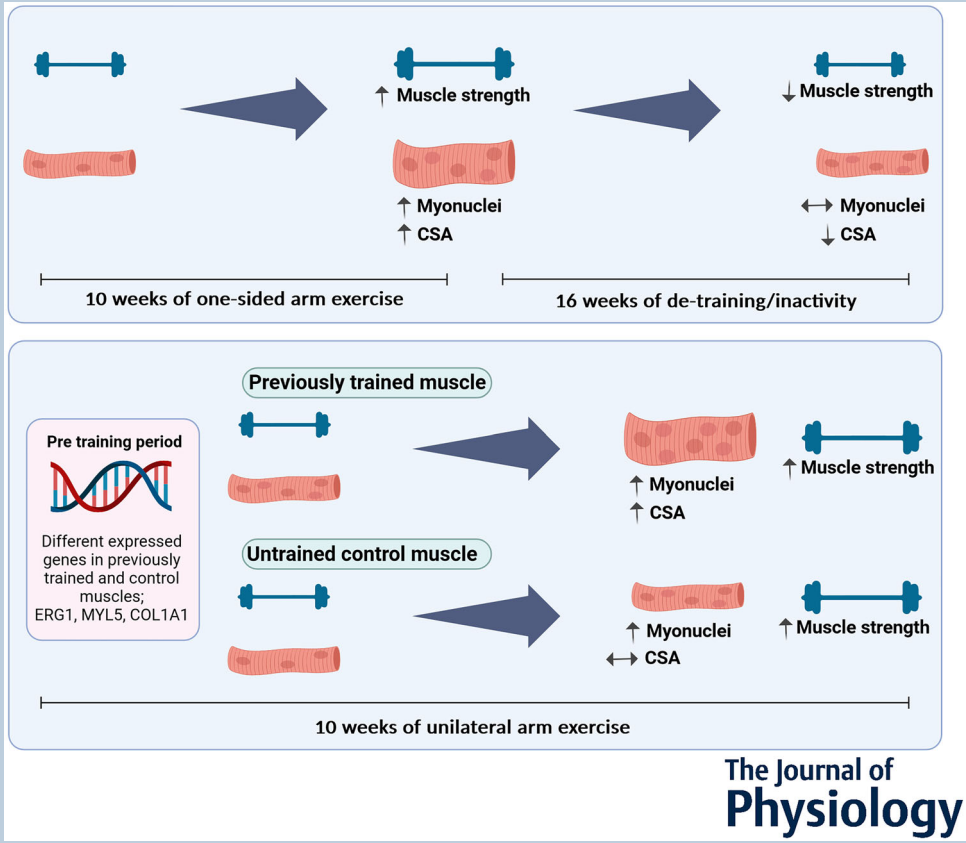

Abstract The objective of this work was to investigate myonuclear permanence and transcriptional regulation as mechanisms for cellular muscle memory after strength training in humans. Twelve untrained men and women performed 10 weeks of unilateral elbow-flexor strength training followed by 16 weeks of de-training. Thereafter, 10 weeks' re-training was conducted with both arms: the previously trained arm and the contralateral untrained control arm. Muscle biopsies were taken

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from the trained arm before and after both training periods and from the control arm before and after re-training. Muscle biopsies were analysed for fibre cross-sectional area (fCSA), myonuclei and global transcriptomics (RNA sequencing). During the first training period, myonuclei increased in type 1 (13 ± 17%) and type 2 (33 ± 23%) fibres together with a 30 ± 43% non-significant increase in mixed fibre fCSA ($P = 0.069$). Following de-training, fCSA decreased in both fibre types, whereas myonuclei were maintained, resulting in 33% higher myonuclear number in previously trained *vs*. control muscle in type 2 fibres. Furthermore, in the previously trained muscle, three differentially expressed genes (DEGs; *EGR1*, *MYL5* and *COL1A1*) were observed. Following re-training, the previously trained muscle showed larger type 2 fCSA compared to the control ($P = 0.035$). However, delta change in type 2 fCSA was not different between muscles. Gene expression was more dramatically changed in the control arm (1338 DEGs) than in the previously trained arm (822 DEGs). The sustained higher number of myonuclei in the previously trained muscle confirms myonuclear accretion and permanence in humans. Nevertheless, because of the unclear effect on the subsequent hypertrophy with re-training, the physiological benefit remains to be determined.

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Abstract figure legend During unilateral training, muscle strength and fibre cross sectional area (CSA) and myonuclei increased. During de-training muscle strength and fibre CSA decreased while myonuclei remained unchanged. In addition to differences in gene expression, differences in myonuclei formed the basis of a study of muscle memory in humans. During a re-training with unilateral arm exercise, both arms (previously trained and untrained control) increased muscle strength and myonuclei, but only the previously trained arm increased fibre CSA during re-training.

Key points

- Muscle memory is a cellular mechanism that describes the capacity of skeletal muscle fibres to respond differently to training stimuli if the stimuli have been previously encountered.
- This study overcomes past methodological limitations related to the choice of muscles and analytical procedures.
- We show that myonuclear number is increased after strength training and maintained during de-training.
- Increased myonuclear number and differentially expressed genes related to muscle performance and development in the previously trained muscle did not translate into a clearly superior responses during re-training. Because of the unclear effect on the subsequent hypertrophy and muscle strength gain with re-training, the physiological benefit remains to be determined.

Introduction

Muscle memory as a cellular mechanism describes the capacity of skeletal muscle fibres to respond differently to environmental stimuli in an adaptive manner if the stimuli have been previously encountered. In the field of strength training and muscle hypertrophy research,

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muscle memory refers to the phenomenon where previously trained muscles re-gain muscle size and strength faster than exercise-naïve muscles after a period of disuse (Gundersen, 2016). The two main mechanisms suggested for the faster muscle growth with re-training are permanent myonuclear accretion and epigenetic modifications (Egner et al., 2013; Seaborne et al., 2018; Snijders et al., 2020).

Staron et al. (1991) reported the first indications of a cellular muscle memory effect in humans, observing that the strength and hypertrophy gained after 20 weeks of training were re-gained faster with only 6 weeks of re-training after a de-training period of 30–32 weeks. Similar results for strength gains were observed in elderly men that were re-trained after 12 weeks of de-training (Taaffe & Marcus, 1997). Intriguingly, muscle hypertrophy in the re-training period in the latter study was non-significant, indicating that the retention of strength to a large part was dependent on neural adaptations. These results were recently confirmed in another training–de-training intervention in elderly (Blocquiaux et al., 2020). Consequently, memory mechanisms underlying the faster increase in strength in human studies seem to include long-lasting neural adaptations while it has been more challenging to document any cellular mechanisms such as myonuclear accretion and permanence.

The memory effect of permanent myonuclear accretion on faster muscle growth was first shown in mice after testosterone supplementation (Egner et al., 2013), and later confirmed with resistance training in rats (Eftestol et al., 2022; Lee et al., 2018). In response to testosterone and training induced hypertrophy, the increase in fibre cross sectional area (fCSA) was accompanied by increased number of myonuclei (Eftestol et al., 2022; Egner et al., 2013; Lee et al., 2018). In these studies, the newly incorporated myonuclei were retained despite the fCSA decreasing back to baseline levels, thus suggesting that myonuclei may be permanent and not removed when adult muscle undergoes atrophy from unloading. The positive memory effect was subsequently observed by the introduction of an overload or training protocol where the muscles enriched in myonuclei displayed larger and faster hypertrophy compared to the control muscles. This supports the hypothesis that an elevated number of myonuclei is a possible mechanism for muscle memory, mainly because the number of myonuclei is a limiting factor for the regulation of fibre size (Hansson et al., 2020). However, this mechanistic explanation has been challenged, with some studies reporting no changes or the opposite, e.g. a reduction in myonuclear number after de-training in older men (Blocquiaux et al., 2020; Snijders et al., 2019) and in mice (Dungan et al., 2019; Murach et al., 2020).

As myonuclear accretion is not consistently observed in human muscle following periods of muscle loading it has previously been challenging to address whether re-training after a period of de-training results in faster muscle growth, i.e. a muscle memory effect (Snijders et al., 2020). In our previous study (Psilander et al. 2019), we aimed to investigate the theory of muscle memory by myonuclear accretion and permanence in human vastus lateralis muscle. The training protocol was designed to optimize increases in myonuclear number by combining heavy-load resistance exercise with short periods of high frequency low-load blood flow restricted resistance exercise, which has been shown to result in large and rapid increases in myonuclear number (Nielsen et al., 2012). However, the training protocol employed in the study by Psilander et al. (2019) failed to induce significant elevations in myonuclear number and consequently did not allow for an investigation of a myonuclei-based muscle memory effect. Furthermore, the exercise intervention targeted knee-extensor muscles that are weight-bearing and used in daily activities. The fact that the participants continued with their normal activities during the de-training period might have contributed to the limited de-training effect observed. In contrast, upper body muscles are less used in daily activities and are therefore easier to manipulate by exercise interventions and importantly, at least in theory, easier to detrain. In the present study we therefore chose a training–de-training–re-training intervention targeting elbow flexor muscles and a within-subject design in order to eliminate the inter-individual variation in training response.

While it has been challenging to confirm muscle memory by permanent myonuclear accretion in humans, the first evidence for muscle memory by epigenetic modifications was indicated in a training study in humans by changes in DNA methylation status (Seaborne et al., 2018). Seven weeks of strength training increased muscle mass and this was paralleled by increased hypomethylation of CpG islands in promotor regions in the trained muscle. For certain genes, the hypomethylated signature was maintained during subsequent de-training while other adaptations such as muscle mass returned to baseline. Importantly, the subsequent increase in muscle mass was larger in the re-training period compared to the first training. Also, in some genes involved in regulation of muscle mass, hypomethylation was further enhanced during re-training, suggesting that these genes retained an epigenetic memory of the previous training. The evidence for muscle memory by epigenetic modifications or transcriptional regulation was later supported by observations in both animals (Wen et al., 2021b) and humans (Moberg et al., 2020). Furthermore, a recent meta-study utilizing 181 samples identified a large number of hypo- and hyper-methylated genes following chronic resistance exercise (Turner et al., 2019). However, conflicting evidence exists regarding the role of the epigenome in

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muscle memory with another previous paper reporting no effect of previous training in a unilateral leg endurance training study (Lindholm et al., 2016). These contradictory results may, however, be related to the different training protocols indicating less memory effects after endurance training than strength training, or challenges related to the use of weight-bearing leg muscle for some of these investigations.

In summary, it has been challenging to verify muscle memory by nuclear accretion and permanence in human studies and the evidence for muscle memory by epigenetic modifications is still scarce. We believe that one limitation in human studies is the focus on leg extensor muscles as these weight-bearing and daily used muscles adapt less to training interventions and are more difficult to detrain than upper body muscles. Further, because the exercise responses in humans are more variable than in genetically homogeneous rodents, between-group comparisons are challenging with a low number of participants. Although several mechanisms have been suggested, we chose to focus on mechanisms related to myonuclear accretion and permanence and long-lasting differences in the transcriptome to further elucidate the mechanisms involved in muscle memory. Therefore, the present study had four different and separate research questions. By using a training–de-training–re-training study targeting elbow flexor muscles in one arm and using the contralateral arm as the control, we aimed to answer these questions: Is myonuclear number increased in the first period of strength training? Is the increased myonuclear number maintained during de-training? Are there any differences in the transcriptional profile between the previously trained compared to the untrained muscle before the re-training period? And lastly, will a higher number of myonuclei and possible differences in the transcriptome result in greater hypertrophy and muscle strength gains in the previously trained compared to the untrained muscle in the re-training period?

Methods

Participants and ethical approval

Twelve healthy subjects (24 ± 3 years old, 72.7 ± 17.8 kg body mass, 172 ± 9 cm body height, 24.3 ± 3.6 kg/m^2 body mass index, 8 females, 4 males) were recruited to the study. Inclusion criteria were that they were healthy and had never performed strength training or any other muscle hypertrophic training (e.g. rowing) of their elbow flexors (biceps brachii and brachialis). All participants were given written and oral information about the project and potential negative effects of participation before they gave their written informed consent to participate in the study. The study was approved by The Regional Committee For Medical Research Ethics South East Norway (ref. 2018/209) and the study conformed to the standards set by the *Declaration of Helsinki*, except for registration in a database.

Experimental design

The study was conducted over a total of 38 weeks consisting of two training periods of 10 weeks separated by a 16-week de-training period that consisted of no training or physical activity involving the elbow flexors (see Fig. 1). The first training period served as a primer period to the main experimental period. Before the first training period, one arm was randomized to training stratified by dominant/non-dominant arm, while the other arm would not be trained, thereby serving as intra-individual control muscle/arm (unilateral training). After 10 weeks of training, the first training period was followed by a 16-week de-training period. During de-training, the subjects did not train their arms or perform any other activity that was considered training and involvement of the elbow flexors. This was emphasized and repeatedly instructed before the start of the de-training and followed up midway through the period by an e-mail or SMS, and when the subjects had completed the de-training they were asked about their activity level and any training that could have affected the elbow flexors. All subjects confirmed that they had not carried out any activity that could influence the goal of the de-training. The de-training period was then followed by a new training period of 10 weeks. During the re-training period, both the previously trained and the untrained control arm were trained (unilateral training) for 10 weeks using the same training program as during the first training period (described in detail later). The week prior to (pre time point) and the week after (post time point) both training periods all subjects completed familiarization with the strength test and training, followed by strength testing, measurements of muscle thickness and muscle biopsies as described in detail later. During the first weeks of the two training periods, all participants had a 24 h recall interview to map their dietary status with focus on energy and protein intake. Data from the recall interview were analysed using an online diet tool (https://www.kostholdsplanleggeren.no/, Norwegian Directorate of Health and the Norwegian Food Safety Authority). If a participant dietary status was not within the recommendation for optimal muscle growth, e.g. protein intake less than 1.4 g/kg/day (Jäger et al., 2017), they were instructed to make specific changes to their diet to meet the recommendations (e.g. increase energy and/or protein intake).

Training protocol

The training program used during the two training periods was unilateral and identical, and the same for both

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arms during re-training, except for the loads, which were adjusted so both arms exercised with the same intensity. The training was gradually increased over the training periods with variation in both exercise loads and exercises (a mix of linear and daily undulating periodization). During the two first weeks of both training periods (week 1–2), strength exercise was performed twice per week and three times per week after that (week 3–10) with one to two exercises targeting the arm flexors. Loads ranged from the heaviest loads that could be lifted for 6–12 repetitions maximum (6–12RM loads) and two to six sets were conducted per training session (5–18 sets per week). When the subjects did three training sessions per week, two were performed to failure and one with submaximal loads (2–3 reps in reserve by set termination). All training sessions were supervised to ensure that the training was performed according to the prescribed regimen and with high quality.

Strength testing

Maximal strength of the elbow flexors was determined by a 1 repetition maximum (1RM) test using a cable apparatus sitting on a preacher bench. After a brief general warm-up at low intensity using a rowing machine, the subjects performed three to four warm-up sets at 10, 7, 5, and 3 repetitions with increasing but submaximal loads. 1RM was defined as the maximal load the subjects could lift from a flexed position to full extension and back to the flexed position. Exercise execution was closely monitored, and the subjects were not allowed to help with the other arm or rotate their torso or shoulder during the lift. The subjects were given verbal encouragement during the test. Three to four minutes of rest was given between each set during the warm-up and the 1RM test (if several tries were needed). The testing procedure was standardized and performed identically for all 1RM tests.

Muscle thickness

Muscle thickness of the elbow flexors (biceps brachii and brachialis) was measured using ultrasound (L12-5, Philips HD11XE 5−12 MHz, Bothell, WA, USA or 9L-D, GE Healthcare Logiq E10, GE Vingmed Ultrasound, Horten, Norway), while the subjects were lying in a supine position. All measurements of muscle thickness were performed before the 1RM test and muscle biopsies, and at least 72 h after any training session. Ultrasound images were captured from the mid portion of the belly of biceps brachii and brachialis in separate scans, with the probe placed gently on the skin in the transverse plane. To ensure reliable scan repositioning between time points, anatomical features such as blood vessels and connective tissue were identified from baseline images and used as references in subsequent scans. Ultrasound images were analysed using Fiji ImageJ 1.5-n (Wayne Rasband, National Institutes of Health, Bethesda, MD, USA).

Muscle biopsies

Muscle biopsies were sampled under local anaesthesia (xylocain adrenalin, 10 mg/ml + 5 μg/ml, AstraZeneca, PLC, London, UK) from the mid-portion of biceps brachii. Before and after TP1, muscle biopsies were only sampled from the exercising arm, while biopsies from both arms were collected before and after TP2. The biopsies were sampled at least 72 h after any training session or strength testing using a Bergström needle biopsy with suction, obtaining approximately 50–100 mg of muscle tissue. The sample(s) were quickly washed in

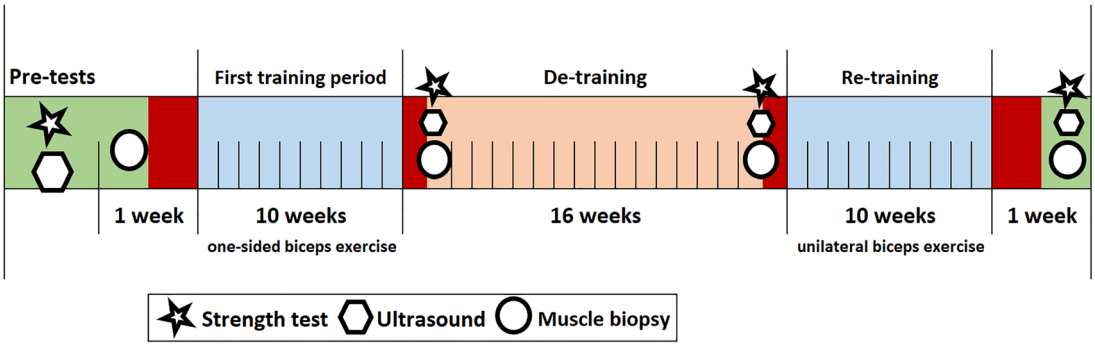

Figure 1. Overview of testing, training, and de-training during the 38-week study period
Strength tests and ultrasound measurements were performed before and after the first training period, and before and after the re-training period for both arms. Muscle biopsies was collected before (week 0; pre) and after (week 10; post) the first training period from the arm that was trained, and from both arms before (week 26; pre) and after (week 36; post) re-training. Weeks shown in green represents test weeks not part of the 36-week intervention period; rest periods separating training and tests are shown in red; training periods are shown in blue; and the de-training period is shown in pink. The star, hexagon, and circle illustrate the time point in which the respective tests were performed.

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physiological saline, and cleaned free from fat, connective tissue and blood. Muscle tissue designated for RNA sequencing analysis (10–20 mg) was gently dried free from physiological saline and immediately snap frozen in liquid nitrogen. The remaining tissue was mounted in embedding medium (KMA-0100-00A, OCT Embedding Matrix, CellPath, Newtown, Powys, UK) before being frozen in 2-methylbutane cooled down to its freezing point in liquid nitrogen. All samples were stored at −80°C until analysed within 6 months.

RNA extraction and sequencing

RNA extraction from 10–20 mg of snap-frozen skeletal muscle tissue was performed using the phenol based TRIzol method (15596018, Thermo Fisher Scientific, Waltham, MA, USA), quantified and quality checked using the 2100 Bioanalyzer System (Agilent Technologies, Santa Clara, CA, USA). Libraries were prepared by poly-A selection (TruSeq mRNA, Illumina, San Diego, CA, USA) and multiplexed at the National Genomics Infrastructure Sweden. RNA sequencing was performed paired end (2 × 150 bp) in two lanes with a S4 flow cell using the NovaSeq 6000 sequencer (Illumina).

Preparation and analysis of frozen sections

Muscle tissue was sectioned at 8 μm thickness at −20°C using a microtome (CM1860 UV, Leica Microsystems, Nussloch, Germany), mounted on microscope slides (Superfrost Plus, J1800AMNZ, Thermo Fisher Scientific) and air dried and stored at −80°C until further analysis. After being thawed, sections were blocked in 1% BSA (A4503, Sigma-Aldrich, St Louis, MO, USA) in phosphate-buffered saline (PBS) with Tween (-T) (524650, Calbiochem, Merck, Darmstadt, Germany) containing 0.05% Tween-20 (437082Q, VWR, Radnor, PA, USA) for 60 min at room temperature before incubation with anti-dystrophin (ab15277, Abcam, Cambridge, UK) and anti-myosin heavy chain 1 (BA-D5, Developmental Studies Hybridoma Bank (DSHB), Iowa City, IA, USA; deposited to the DSHB by S. Schiaffino) antibodies diluted in 1% BSA overnight at 4°C. After the overnight incubation the sections were washed with gentle agitation 3x for 10 min in PBS-T before incubation in secondary antibodies (A11005 and A11001, Thermo Fisher Scientific) for 60 min at room temperature. After incubation the sections were then washed again 3 × 10 min in PBS-T, before covered with a coverslip (0107222, No. 1,5H tol. ± 5 μm, Marienfeld, Lauda-Könighofen, Germany) and mounted (ProLong Gold Antifade Reagent with 4′,6-diamidino-2-phenylindole (DAPI), P36935, Thermo Fisher Scientific). Sections analysed for myonuclei were prepared according to a modified protocol described by Winje et al. (2018). Briefly, sections were blocked with 2% BSA in PBS for 30 min at room temperature before incubated with anti-PCM1 (HPA023370, Sigma-Aldrich) and anti-dystrophin (MANDYS8(8H11), Developmental Studies Hybridoma Bank (DSHB); deposited to the DSHB by G. E. Morris) diluted in 5% BSA in PBS with 0.2% Igepal CA-630 (I8896, Sigma-Aldrich) overnight at 4°C. After 3×10 min washes in PBS with gentle agitation, the sections were incubated with secondary antibodies (A11008, Thermo Fisher Scientific, and CF594, 20110, Biotium, Fremont, CA, USA) diluted in 2% BSA in PBS for 60 min at room temperature. Following incubation, the sections were washed for 3×10 min in PBS before being coverslipped and mounted as described. All sections were left to cure overnight at room temperature before they were stored at −20°C until being further analysed.

Sections were imaged using a microscope (BX61, Olympus, Tokyo, Japan) with a mounted camera (DP72, Olympus) using a ×4, 0.13 NA air objective (UPlanFL N, Olympus) for fibre CSA and myosin heavy chain analyses or a ×10, 0.30 NA (UPlanFL N) air objective for the myonuclear analyses. Individual muscle fibre CSAs were measured and analysed using TEMA software (Checkvision, Hadsund, Denmark). CSA analyses included 264 ± 138 (mean ± standard deviation; range 99–749) type 1 fibres and 421 ± 167 (range 125–954) type 2 fibres from each sample.

By the criteria described by Winje et al. (2018), myonuclei were manually analysed using Fiji software (Schindelin et al., 2012) and related to their respective fibre type using the image analysed for CSA and fibre type. A total of 50 type 1 and 50 type 2 fibres from each sample were included in the analyses. In addition, we quantified the percentage of myonuclei (PCM1 positive DAPI labelled nuclei) of total nuclei (DAPI labelled nuclei), leaving out areas between fascicles, which were excluded from the analyses. The number of nuclei from each sample included in the analyses was 871 ± 266 (mean ± standard deviation; range 397–1322) per time point.

Statistics and bioinformatic analysis

Based on the changes in fCSA reported by Bamman et al. (2007) a sample size calculation was performed using G*power 3.1 (Faul et al., 2007). Since the study by Bamman et al. had lower variation in the changes in fCSA than what we would expect in our study, we chose to double the standard deviation reported in our sample size calculation. An α level of 0.05 and power 0.8 was used. The calculation recommended a total of nine participants for the paired comparisons with our within-subject design. The statistical analysis of physiological and histological data was performed using GraphPad Prism 10.1.2. (GraphPad Software, Boston, MA, USA). Student's two-sided paired sample *t* test was

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performed analysing for changes and differences between muscles and arms. Correlations between theoretically related variables were tested using the Pearson correlation coefficient. Bioinformatic analyses were performed with R version 3.6.0 (R Foundation for Statistical Computing, Vienna, Austria). For differential gene expression analysis, limma (Ritchie et al., 2015) and edgeR (Robinson et al., 2010) packages were used. Gene set enrichment analysis was performed with the fgsea package (Korotkevich et al., 2021) using the gene ontology libraries for biological process, molecular function and cellular compartment (Ashburner et al., 2000). For over-representation analysis, the PANTHER classification system was used (Mi & Thomas, 2009; Mi et al., 2021). Statistical analyses were performed separately for the three periods investigated with the aim of answering the research questions. Data are reported as means ± standard deviations, unless stated otherwise, and/or individual data points. Statistical significance was set at $P < 0.05$ unless stated otherwise.

Results

Progression in training load

In the first training period, the 10RM training load used in the main exercise increased from 4.9 ± 2.3 kg in the first week to 8.3 ± 2.9 kg in the last week (increase of 79 ± 32% in 10RM training load, $P < 0.001$). In the re-training period, the training load was higher in the previously trained arm compared to the control arm ($P = 0.004$). In the same training period, the 10RM training load increased for the previously trained arm from 6.5 ± 2.4 to 11.4 ± 3.5 kg (increase of 81 ± 27% in 10RM training load; $P < 0.001$) and from 5.6 ± 2.2 to 10.3 ± 3.0 kg in the control arm (increase of 94 ± 32% in 10RM training load; $P < 0.001$). At the last training week, the previously trained arm trained with a higher training load compared to the control arm ($P = 0.001$).

Dietary status

During the first training period the 24 h dietary recall showed that the subjects had an energy intake of 2642 ± 510 kcal (range 1521–3150 kcal). This was not different from the energy intake during the re-training period ($P = 0.089$), with an energy intake of 3115 ± 1086 kcal (range 1318–5319 kcal). The average protein intake was 1.64 g per kg body weight (range 0.89–2.46 g per kg) during the first training period, which was not different ($P = 0.358$) from the re-training period where the protein intake was 1.81 g per kg body weight (range 0.62–3.09 g per kg body weight). Three subjects had a protein intake under the recommended daily intake for optimal muscle growth. All three subjects increased their protein intake at the 24 h recall during re-training, but two of them were still under the recommended intake.

Histological results from the previously trained muscle

Herein, only the results from the previously trained muscle from the first training period, de-training and re-training will be presented.

Fibre cross sectional area. Overall, mixed fibre fCSA had a non-significant increase of 30 ± 43% ($P = 0.069$) in the first training period; when analysed separately the changes were not significant for type 1 (21 ± 38%; 3527 ± 684 *vs.* 4251 ± 1653 μm^2, $P = 0.181$; Fig. 2*B*) or for type 2 fibres (34 ± 48%; 4029 ± 1026 *vs.* 5252 ± 2270 μm^2, $P = 0.112$; Fig. 2*B*). In the de-training period, a decrease in mixed fibre fCSA was observed (−21 ± 12%; 4920 ± 2051 *vs.* 3764 ± 1159 μm^2, $P = 0.011$), which was reflected by a decrease in both fibre types (−17 ± 12%, 4251 ± 1653 *vs.* 3405 ± 931 μm^2, $P = 0.012$, and −23 ± 13%, 5252 ± 2270 *vs.* 3897 ± 1275 μm^2, $P = 0.01$, for type 1 and 2 respectively; Fig. 2*B* and *C*). In the re-training period, mixed fibre fCSA increased by 22 ± 26% (3764 ± 1159 *vs.* 4512 ± 1293 μm^2, $P = 0.034$). Type 2 fCSA increased (30 ± 29%; 3897 ± 1275 *vs.* 4992 ± 1661 μm^2, $P = 0.015$; Fig. 2*C*) whereas there was no change in type 1 fCSA (14 ± 23%; 3405 ± 931 *vs.* 3758 ± 722 μm^2, $P = 0.168$; Fig. 2*B*).

Number of myonuclei. The number of myonuclei per fibre increased in the first training period for both type 1 (13 ± 17%; 2.0 ± 0.3 *vs.* 2.3 ± 0.4 myonuclei per fibre, $P = 0.017$; Fig. 3*B* and *C*) and type 2 fibres (34 ± 22%; 2.4 ± 0.5 *vs.* 3.3 ± 0.7 myonuclei per fibre, $P < 0.001$; Fig. 3*C*). In the de-training period, there was no change in the number of myonuclei either in type 1 (−1 ± 7%, $P = 0.752$; Fig. 3*B*) or in type 2 fibres (−2 ± 5%, $P = 0.276$; Fig. 3*C*). In the re-training period, myonuclear number further increased by 8 ± 6% (2.2 ± 0.4 *vs.* 2.3 ± 0.4 myonuclei per fibre, $P = 0.002$; Fig. 3*B*) in type 1 and by 15 ± 7% (3.2 ± 0.6 *vs.* 3.6 ± 0.7 myonuclei per fibre, $P < 0.001$; Fig. 3*C*) in type 2 fibres.

The proportion of myonuclei expressed as the percentage of total nuclei decreased in the first training period (from 69 ± 6% *vs.* 61 ± 6%, $P = 0.001$; shown in Fig. 6*C*). No changes in proportion of myonuclei was observed in the de-training period in the previously trained muscle (from 61 ± 6% to 65 ± 5%, $P = 0.072$; shown in Fig. 6*C*), and the same observation was seen after the re-training period (from 65 ± 5% to 62 ± 4%, $P = 0.09$; shown in Fig. 6*C*).

Myonuclear domain. Type 1 and 2 myonuclear domain size (fCSA/myonuclear number per fibre) remained

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unchanged in response to the first training period. During the de-training period the myonuclear domain decreased in both the type 1 (1919 ± 462 to 1574 ± 258 μm^2, $P = 0.009$) and type 2 fibres (1595 ± 390 to 1224 ± 251 μm^2, $P = 0.003$). During the re-training period no change in myonuclear domain was observed in both types of fibre types.

Muscle thickness and strength in the previously trained and control arm

Herein, results from both arms from the first training period, de-training, and re-training will be presented.

Muscle thickness. Muscle thickness measured with ultrasound increased in the first training period in the trained arm by 13 ± 5% (29.0 ± 5.3 *vs.* 32.6 ± 5.6 mm, $P < 0.001$; Fig. 4), followed by a decrease by 8 ± 4% (32.6 ± 5.6 *vs.* 30.3 ± 5.9 mm, $P < 0.001$; Fig. 4) in the de-training period. The control arm had no changes in muscle thickness during the same two periods (Fig. 4). At the end of the first training period, the trained arm had a higher muscle thickness compared to the control arm ($P = 0.003$). At the start of the re-training period, there were no differences in muscle thickness between the two arms ($P = 0.327$). Over this training period, both arms increased muscle thickness by 13 ± 5% (30.3 ± 5.9 *vs.*

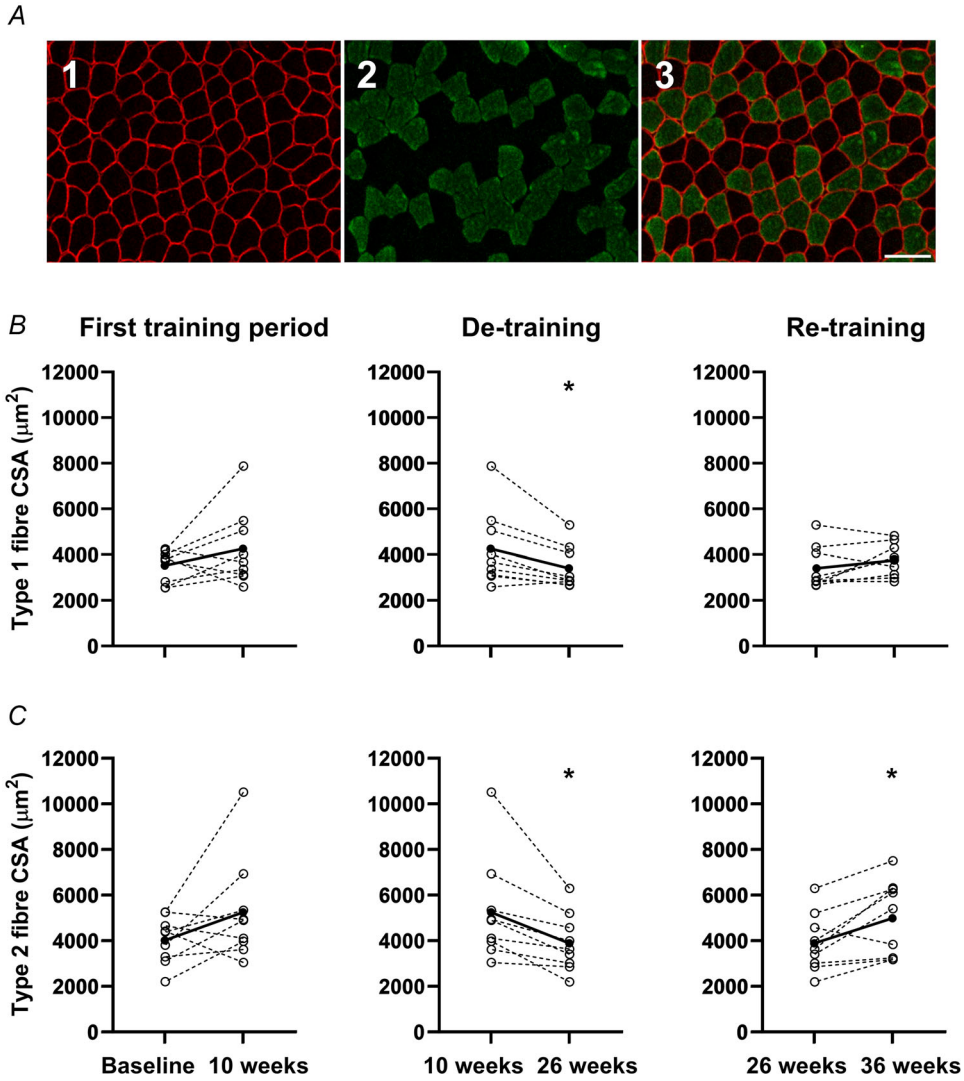

Figure 2. Muscle fibre CSA from the trained muscle
A, representative micrographs of histological sections. 1, staining against dystrophin; 2, staining against MHC1; 3, merged picture (picture 1+2). Scale bar in lower right corner = 100 μm. *B* and *C*, muscle fibre cross sectional area (CSA) in the 'trained arm' for type 1 (*B*) and type 2 (*C*) fibres in the first training period ($n = 9$; week 0/baseline–10), de-training ($n = 9$; week 10–26) and re-training period ($n = 9$; week 26–36). Open circles and dashed lines represent the results from each participant ($n = 9$). Filled circles and continuous line represent mean values. *Significant difference ($P < 0.05$) from the first time point within the same experimental period.

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34.3 ± 7.0 mm, $P < 0.001$; Fig. 4) in the previously trained arm and by 15 ± 5% (30.6 ± 5.8 *vs.* 34.9 ± 6.3 mm, $P < 0.001$; Fig. 4) in the control arm, which was similar to the increase observed in the first training period of the trained arm. At the end of the training period, there were no differences in muscle thickness between the two arms ($P = 0.304$).

Muscle strength. Muscle strength measured as 1RM elbow flexion, increased in the first training period in the trained arm by 50 ± 26% (9.8 ± 3.8 *vs.* 13.9 ± 4.6 kg, $P < 0.001$; Fig. 5), followed by a decrease by 7 ± 5% (14 ± 4.8 *vs.* 13.3 ± 5.0 kg, $P < 0.001$; Fig. 5) in the de-training period. For the control arm, 1RM muscle strength increased by 9 ± 8% (10 ± 4.2 *vs.* 11 ± 4.1 kg, $P < 0.001$; Fig. 5) in the first training period and was unchanged in the de-training period ($P = 0.554$). Regardless of the increase in 1RM in the control arm, the trained arm was stronger ($P < 0.001$) at the end of the first training period. At the start of the re-training period, the previously trained arm was stronger than the control arm (13.3 ± 5.0 *vs.* 11.2 ± 4.4 kg, $P = 0.004$). Both arms increased muscle strength over the re-training period with the previously trained arm increased by 38 ± 19% (13.3 ±

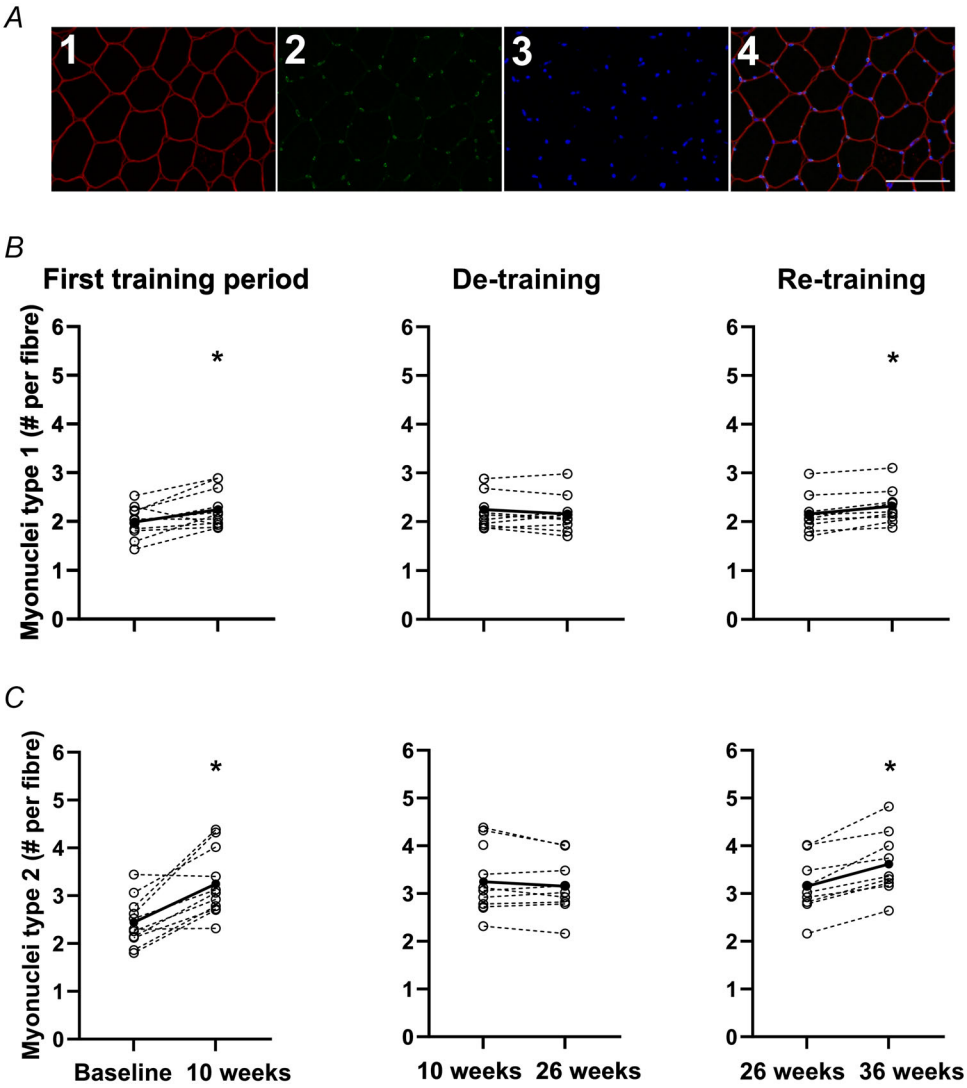

Figure 3. Number of myonuclei in the trained muscle
A, representative micrographs of histological sections. 1, staining against dystrophin; 2, staining against PCM1; 3, DAPI labelling; 4, merged picture (picture 1+2+3). Scale bar in lower right corner = 100 μm. *B* and *C*, number of myonuclei in type 1 (*B*) and type 2 (*C*) fibres in the first training period (n = 11; week 0/baseline-10), de-training (n = 9; week 10–26) and re-training period (n = 9; week 26–36). Number of myonuclei are given as numbers per fibre for each individual participant. Open circles and dashed lines represent the results from each participant. Filled circles and continuous line represent mean values. *Significant difference ($P < 0.05$) from the first time point within the same experimental period.

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5.0 *vs.* 17.8 ± 5.5 kg, $P < 0.001$; Fig. 5) and the control arm increased by 61 ± 30% (11.2 ± 4.4 *vs.* 17.1 ± 4.9 kg, $P < 0.001$; Fig. 5), which resulted in a slightly higher 1RM in the previously trained arm compared to the control arm at the end of this training period ($P = 0.044$). By just analysing the percentage change in muscle strength during the re-training period, the control arm had a larger increase in 1RM compared to the previously trained arm (61 ± 30% *vs.* 38 ± 19%; $P = 0.004$), which was also reflected by a higher delta change in 1RM (4.5 ± 1.4 *vs.* 5.9 ± 1.8 kg, $P = 0.009$) over the re-training period.

Histology and gene expression in the previously trained and control muscle re-training

Herein, results from both muscles from the first training period, de-training and re-training will be presented.

Number of myonuclei. In the previously trained muscle, the number of myonuclei in type 2 fibres was higher at the start of the re-training period compared to the control muscle (3.2 ± 0.6 *vs.* 2.4 ± 0.5, $P = 0.001$; Fig. 6*B*), but not in the type 1 fibres (2.2 ± 0.4 *vs.* 1.9 ± 0.3, $P = 0.078$; Fig. 6*B*). The proportion of myonuclei given as the percentage of all nuclei, was lower in the previously trained muscle compared to the control muscle (65 ± 5 *vs.* 72 ± 4%, $P < 0.001$; Fig. 6*C*) at the start of the re-training period.

Fibre cross sectional area. There was no difference in mixed fibre fCSA between the two muscles at the start of the re-training period ($P = 0.705$), type 1 ($P = 0.247$; Fig. 6*A*) or type 2 fibres ($P = 0.693$; Fig. 6*A*). After the re-training period mixed fibre fCSA had increased in the previously trained muscle (3764 ± 1159 *vs.* 4512 ± 1293 μm^2, $P = 0.034$), with no increase in the control muscle ($P = 0.42$). At the end of the re-training period, the previously trained muscle had higher mixed fibre fCSA compared to the control muscle ($P = 0.021$). However, the delta change over the training period was not different between the two muscles (673 ± 860 *vs.* 203 ± 754 μm^2, $P = 0.225$). No change was observed in fCSA for type 1 fibres for either muscle (Fig. 6*A*) during re-training. For the type 2 fibres, the previously trained muscle increased fCSA by 30 ± 29% ($P = 0.015$) *vs.* 14 ± 25% ($P = 0.173$) in the control muscle (Fig. 6*A*), which resulted in higher fCSA in the type 2 fibres in the previously trained muscle compared to the control

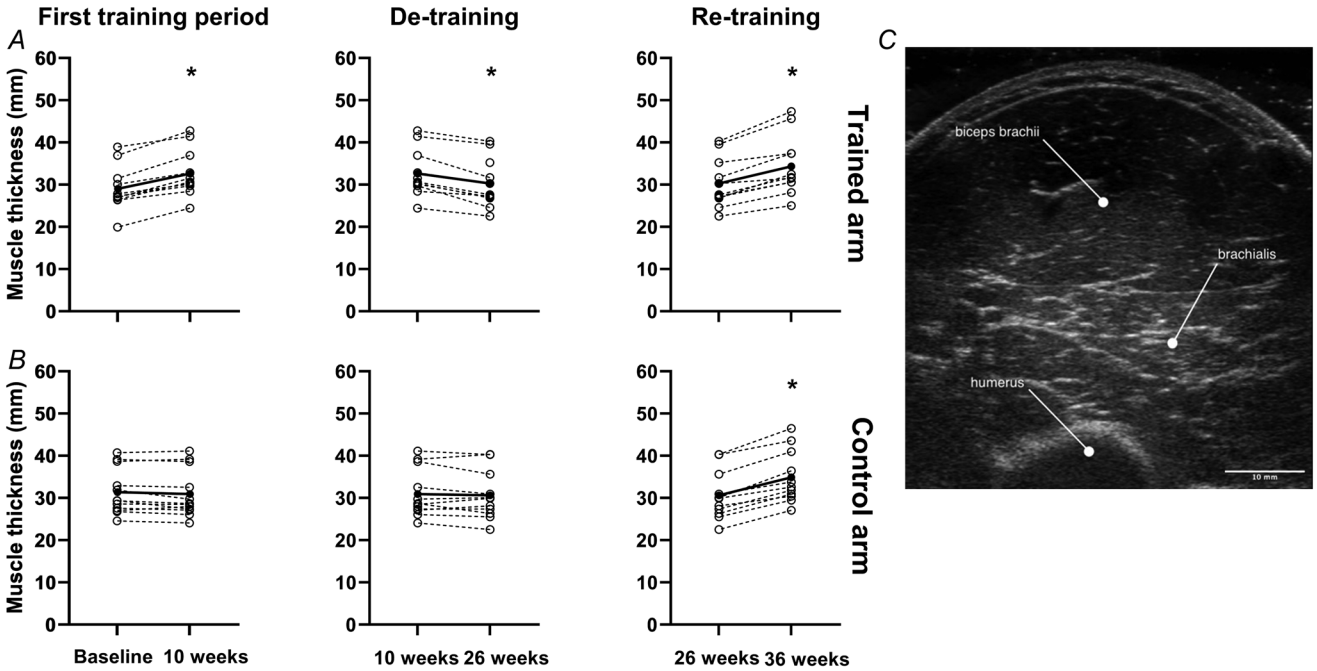

Figure 4. Muscle thickness of biceps brachii and brachialis measured before and after first training period (trained arm only; *n* = 12; week 0/baseline-10), de-training (*n* = 11; week 10–26) and re-training period (*n* = 11; week 26–36)
A, trained arm performing training in both periods. *B*, control arm only performing training in the re-training period. *C*, representative ultrasound image from the biceps brachii and brachialis muscle relative to the humerus. Dotted line denotes the measured muscle thickness of the biceps brachii and brachialis. Scale bar in lower right corner = 10 mm. Muscle thickness given in mm. Open circles and dashed lines represent the results from each participant. Filled circles and continuous line represent mean values. *Significant difference ($P < 0.05$) from the first time point within the same experimental period.

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muscle at the end of the re-training period (4992 ± 1661 *vs.* 4472 ± 1521 μm^2, $P = 0.035$). However, the delta change in fCSA for the type 2 fibres was not different between the two muscles ($P = 0.185$). The number of myonuclei per fibre increased in type 1 fibres in both the previously trained muscle ($P = 0.002$) and the control muscle (1.9 ± 0.3 *vs.* 2.1 ± 0.3 myonuclei per fibre, $P < 0.001$; Fig. 6*B*). But no differences were observed between the two muscles ($P = 0.160$), and the delta change was similar between the two muscles at the end of the re-training period ($P = 0.192$). In the type 2 fibres, myonuclear number increased in both the previously trained muscle ($P < 0.001$) and the control muscle (2.4 ± 0.5 *vs.* 3.1 ± 0.6 myonuclei per fibre, $P < 0.001$; Fig. 6*B*). At the end of the re-training period this resulted in higher myonuclear number in the previously trained muscle ($P = 0.003$). However, the delta change was higher in the control muscle compared to the previously trained muscle (0.7 ± 0.3 *vs.* 0.5 ± 0.2, $P = 0.003$; Fig. 6*B*).

Myonuclear domain. The myonuclear domain was significantly lower in the previously trained compared to the untrained muscle at the start of the re-training period in both type 1 (1574 ± 258 *vs.* 1921 ± 224 μm^2, $P = 0.008$) and type 2 fibres (1224 ± 251 *vs.* 1664 ± 461 μm^2, $P = 0.023$) and remained unchanged in both muscles after the re-training period.

The proportion of myonuclei decreased in the control muscle after the re-training period (72 ± 4 *vs.* 58 ± 3%, $P < 0.001$; Fig. 6*C*), and remained lower in the control muscle compared to the previously trained muscle (58 ± 3 *vs.* 62 ± 4%, $P = 0.049$; Fig. 6*C*).

Correlations. Because of the large individual variation in the changes in fCSA, the correlation between individual myonuclear content before and the corresponding change in fCSA during the re-training period was investigated. In order to normalize myonuclear content to fibre size, myonuclear domain was used in this correlation for both type 1 and 2 fibres. The myonuclear domain before the re-training period correlated with the change in fCSA for the type 1 fibres ($r = -0.59$, $P = 0.009$; Fig. 7), and with a trend for the type 2 fibres ($r = -0.42$, $P = 0.086$; Fig. 7).

Memory effect on global gene expression

At the start of the re-training period there were no differences between muscles in total RNA per mg muscle

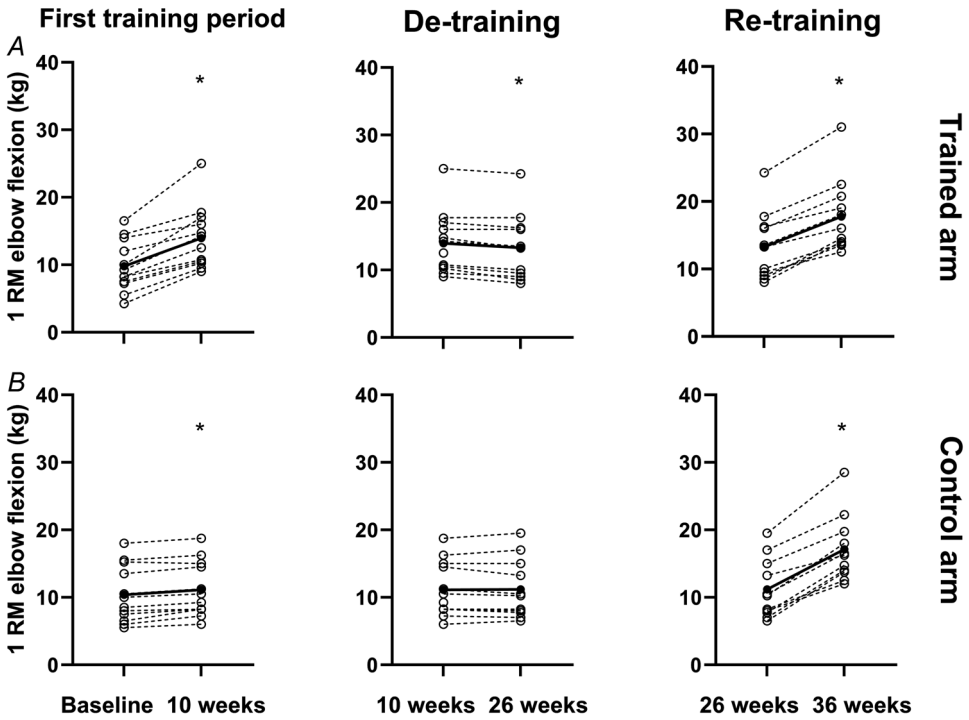

Figure 5. Strength measured as 1 repetition maximum (1RM) in elbow flexion tested before and after first training period (trained arm only; *n* = 12; week 0/baseline-10), de-training (*n* = 11; week 10–26) and re-training period (*n* = 11; week 26–36)
A, trained arm performed training in both periods. *B*, control arm only performed training in the re-training period. 1RM results are given as kg for each individual participant. Open circles and dashed lines represent the results from each participant (*n* = 11–12). Filled circles and continuous line represent mean values. *Significant different ($P < 0.05$) from the first time point within the same experimental period.

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($P = 0.812$). Following the re-training period both the previously trained muscle (11.3 ± 3.2 *vs.* 15.3 ± 3.2 ng/µl RNA per mg muscle, $P = 0.014$) and control muscle (11 ± 1.6 *vs.* 15.5 ± 3.5 ng/µl RNA per mg muscle, $P = 0.004$) increased total RNA. To further investigate the effects of previous training on skeletal muscles, gene expression from before (pre) and after (post) the re-training period was analysed using RNA sequencing. Three genes relevant for skeletal muscle performance or development (*EGR1*, *MYL5* and *COL1A1*; Fig. 8) were differentially expressed between the previously trained and control muscle after the de-training period (adjusted $P = 0.014$, $P = 0.045$ and $P = 0.014$, respectively).

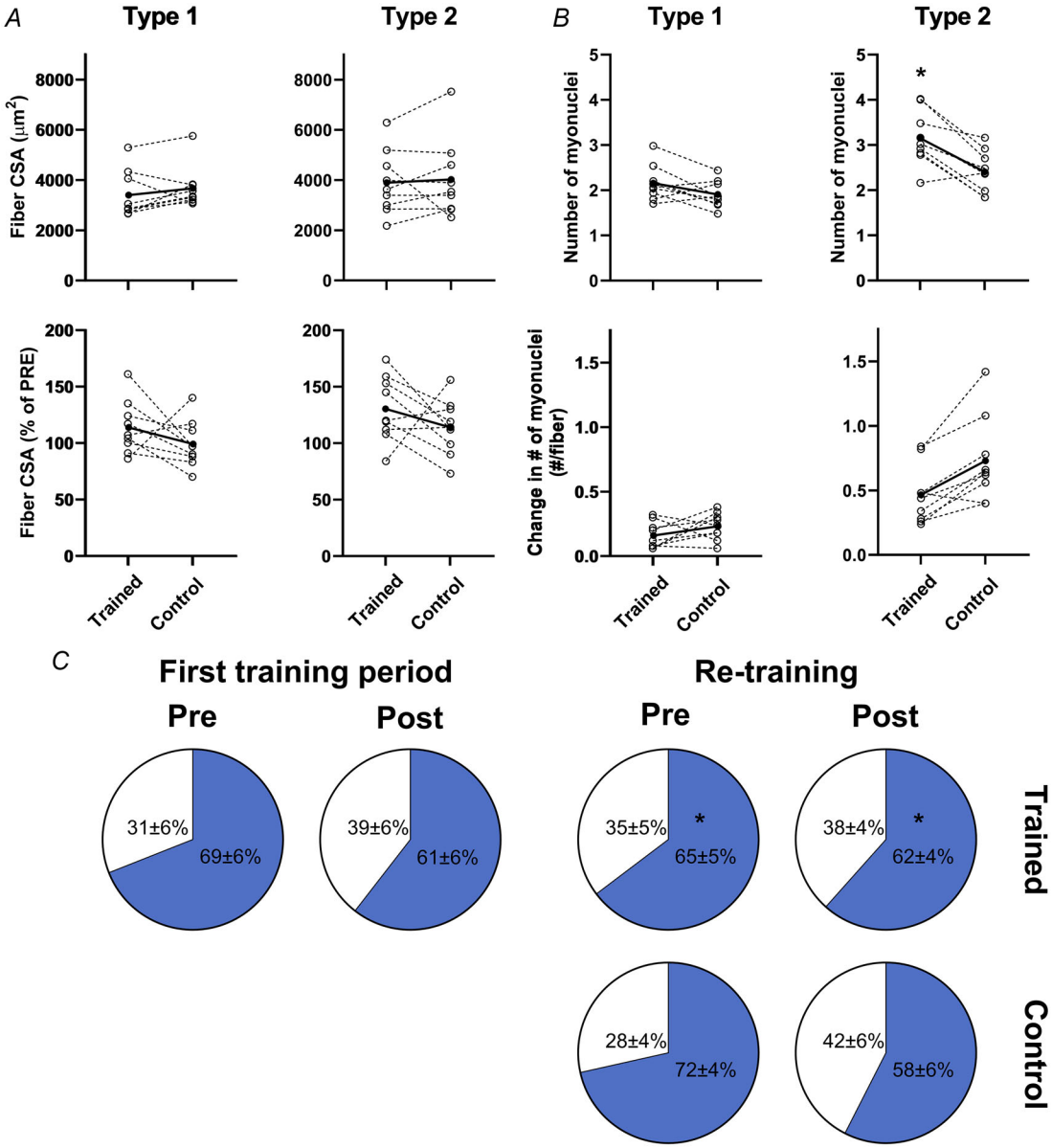

Figure 6. Fibre CSA, myonuclei and percentage of nuclei in the re-training period
A, absolute values for fibre cross sectional area before the re-training period (upper panels) and change in fibre cross sectional area (% of pre re-training values) during the re-training period (lower panels) for the previously trained and control muscles. *B*, absolute values for myonuclear number before the re-training period (upper panels) and change in number of myonuclei (delta number, lower panels) during the re-training period for the previously trained and control muscles. Open circles and dashed lines represent the results from the same participant but from different arms/muscles. Filled circles and continuous line represent mean values. *Significant difference ($P < 0.05$) between previously trained and control. *C*, percentage of all nuclei that are myonuclei (blue) and non-myonuclei (white) in the trained and control muscles before (pre) and after (post) the first training period (trained only; $n = 11$) and re-training (previously trained and control; $n = 9$).

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Subsequently, further differential gene expression analysis revealed large transcriptomic changes after the re-training period. While the control muscle had 1338 genes differentially expressed comparing pre and post re-training (Fig. 9), the previously trained muscle only had 822 differentially expressed genes (DEGs). Of these genes, 587 were shared between the previously trained and the control muscle (Fig. 9). On a global level, following the re-training period, analysis showed no significant DEGs comparing previously trained and control muscle. However, when the three genes of interest identified at the time point before the re-training period were analysed individually, there was a significant difference between the two muscles, with higher *MYL5* expression in the previously trained muscle compared to the control muscle following the re-training period ($P = 0.026$; Fig. 8*B*).

Three groups of genes (exclusive to control muscle, exclusive to the trained muscle or shared), were

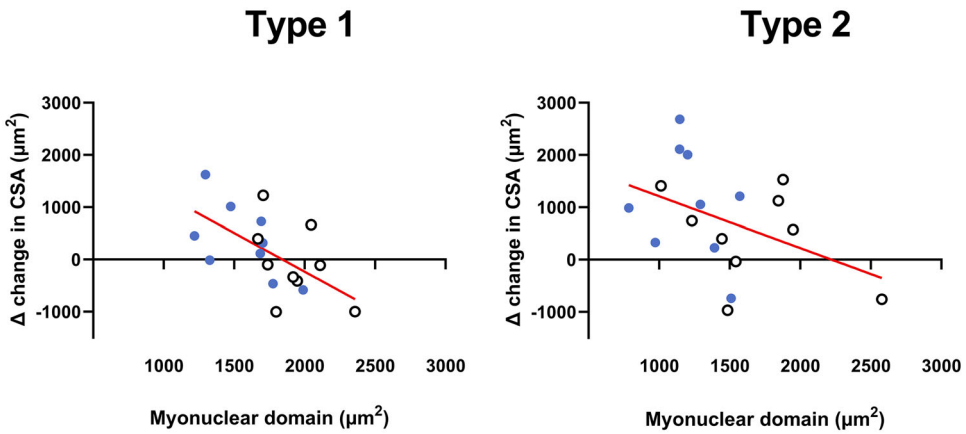

Figure 7. Correlations between delta change in fCSA over the re-training period and myonuclear domain at pre re-training for the type 1 fibres and type 2 fibres
Blue points illustrate individual data points from the previously trained muscle (n = 9), white points illustrate individual data points from the control muscle (n = 9). Red line indicates the linear regression line. Delta change in CSA is expressed as μm^2, and myonuclear domain (fCSA/myonuclei per fibre) is expressed as μm^2.

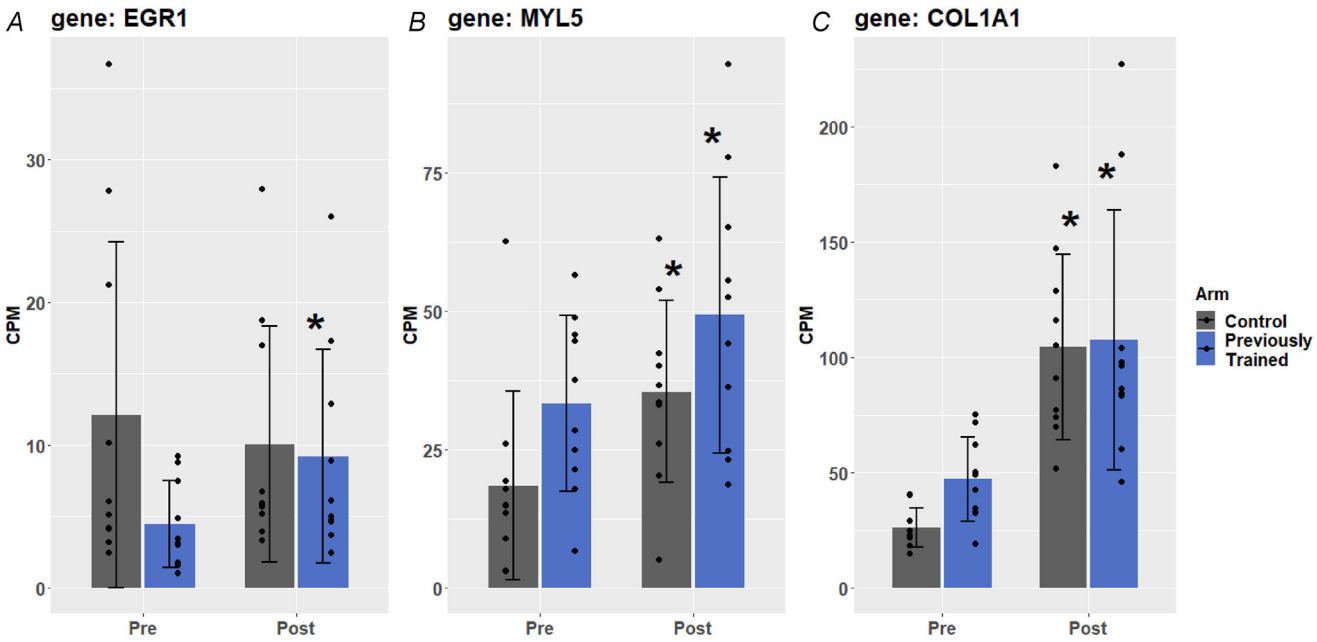

Figure 8. Expression levels of EGR1, MYL5 and COL1A1
Expression levels, expressed and normalised as counts per million (CPM), of the three genes *EGR1* (*A*), *MYL5* (*B*), and *COL1A1* (*C*) differentially expressed in previously trained (blue) *vs*. control (grey) muscles before (pre) and after (post) the re-training period (n = 10) *Significant ($P < 0.05$) different from pre. Error bars represent standard deviation.

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analysed using over-representation analysis using the PANTHER pathway database. Results showed a significant over-representation of genes that are associated with platelet-derived and epidermal growth factor (EGF) pathways (adjusted $P = 0.025$ and $P = 0.036$, respectively), as well as the integrin (cell–cell and extracellular matrix adhesion) pathway (adjusted $P = 0.019$) in genes exclusive to the control muscle. In the list of genes that were common to both arms, cytoskeletal regulation by Rho-GTPase and the integrin signalling pathway were significantly over-represented (adjusted $P = 0.029$ and $P < 0.001$, respectively). No pathways were over-represented in the previously trained muscle-exclusive genes (see Supporting information, Supplement 1).

Further analysis of DEGs between muscles both before and after re-training and between these time points in both muscles was performed using gene set enrichment analysis (GSEA). This analysis revealed a large number of terms (gene sets) related to ribosomal structure and function enriched in the previously trained muscle before the re-training period (Fig. 10*A*). At the time point following the re-training period (post), the balance shifted to a larger number of pathways and terms enriched in the control muscle compared to the previously trained muscle. The terms enriched in the control muscle after training was related to muscle structure and growth such as sprouting angiogenesis, cell matrix adhesion and peptide receptor activity (adjusted $P = 0.004$, $P = 0.004$ and $P = 0.006$, respectively; Fig. 10*B*) and inflammation and immune system, such as IL-10 production, leukocyte migration and inflammatory response (all three adjusted $P = 0.004$) being significant.

Comparing the post time point to pre re-training, there is a large overlap between the previously trained and control muscle in terms enriched following the re-training period. These terms appear related to structural properties of the muscle cell such as collagen trimer, extracellular matrix and basement membrane and the immune system and immune response (Fig. 10*C* and *D*).

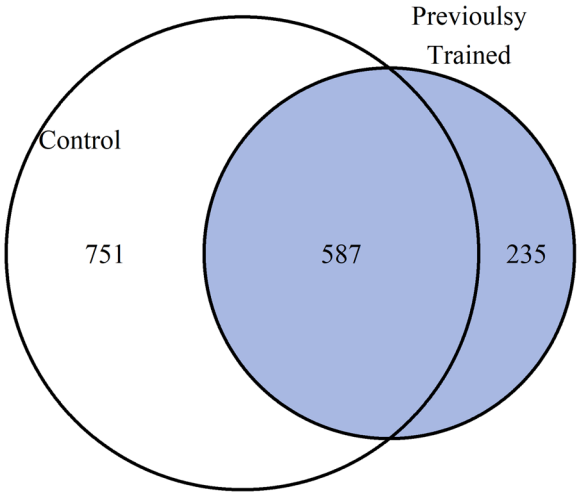

Figure 9.
Number of unique and shared differentially expressed genes comparing post *vs.* pre time point in the re-training period between the trained and control muscle (n = 10).

Summarizing the significant pathways from GSEA of all comparisons, the difference between comparisons of muscles before and after re-training can be further visualized. Both muscles respond to a period of training (longitudinal comparison of previously trained and untrained control muscle) largely in a similar way (Fig. 11*A*). Although the previously trained muscle showed a higher pathway enrichment compared to the control muscle before the re-training period, this dynamic was reversed after the re-training period, with pathways being more enriched in the control muscle. When specifically focusing on gene ontology terms related to the ribosome only, a summary of all member genes of these pathways again shows a similar response shape in the longitudinal perspective (Fig. 11*B*). Directly comparing muscles shows a shift of ribosome-related genes from more highly expressed in the previously trained muscle before (median logFC: 0.22) to more similarly expressed in both muscles after the re-training period (median logFC: −0.07).

Discussion

The aim of the present study was to investigate muscle memory with special focus on myonuclear number and differences in transcriptional profile. Our results clearly showed that myonuclear number was increased in the first training period and maintained during 16 weeks of de-training despite a decrease in fibre sizes and muscle size. Myonuclear permanence during de-training atrophy can serve as a potential mechanism for muscle memory in a subsequent re-training period. Further, RNA sequencing analyses showed that three genes related to skeletal muscle performance and development were differentially regulated in the previously trained compared to the control muscle after de-training, pointing at an additional transcription-regulatory memory mechanism probably related to epigenetic modifications. The increased number of myonuclei and the differentially regulated genes in the previously trained muscle support the hypothesis of muscle memory at the cellular and molecular level. These observations did not, however, translate directly into superior muscle growth or faster increase in muscle strength compared to the control muscle in the re-training period. Consequently, we were not able to either confirm or reject a possible role of muscle memory in humans after the re-training period. Nevertheless, a possible positive role of nuclear accretion, or at least of nuclear density, was supported by the significant correlation between individual myonuclear domains before the re-training

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period and the subsequent degree of muscle hypertrophy in the re-training period.

Myonuclear accretion and permanence

In the first training period, myonuclear number increased in both fibre types. Consequently, the basis for muscle memory by myonuclear accretion was established (Gundersen, 2016; Snijders et al., 2020). Importantly, the increased myonuclear content was maintained during de-training although fCSA of both fibre types decreased and consequently resulted in decreased myonuclear domains. At the start of the re-training period nuclear number was about 33% higher in the type 2 fibres in the previously trained muscle compared to the control muscle whereas the size of the fibres was identical. According to the muscle memory theory, this should improve the capacity for future hypertrophy when exposed to re-training (Egner et al., 2013).

A faster re-gain in muscle strength and/or mass in previously strength trained muscles upon re-training has been shown in several studies in humans (Seaborne et al., 2018; Staron et al., 1991; Taaffe & Marcus, 1997). In our study, we did not, however, observe faster increases in muscle mass or strength in the previously trained arm. In fact, the increase in 1RM elbow flexion strength was larger in the control arm than in the previously trained arm in the retaining period (61 *vs*. 38%). Importantly, the previously trained arm was significantly stronger than the control arm after the de-training period because the

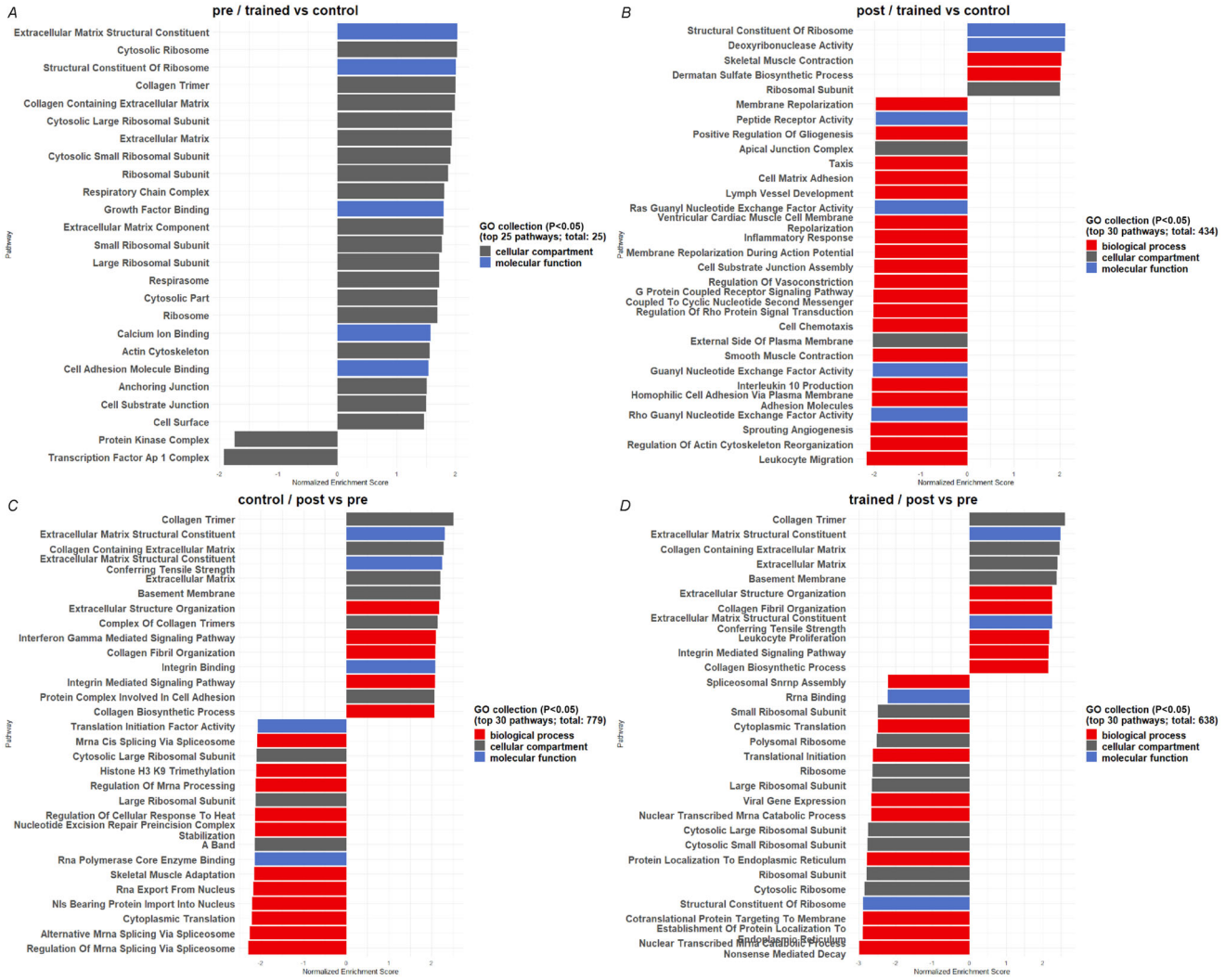

Figure 10. Gene set enrichment analysis
Results of gene set enrichment analysis (GSEA) based on differential gene expression analysis comparing control and previously trained muscle at pre (*A*) and post (*B*) re-training and comparing post *vs*. pre in control (*C*) and previously trained muscle (*D*) (n = 10) Top 30 significant ($P < 0.05$) pathways are shown with the total number of significant pathways in the legend of each panel.

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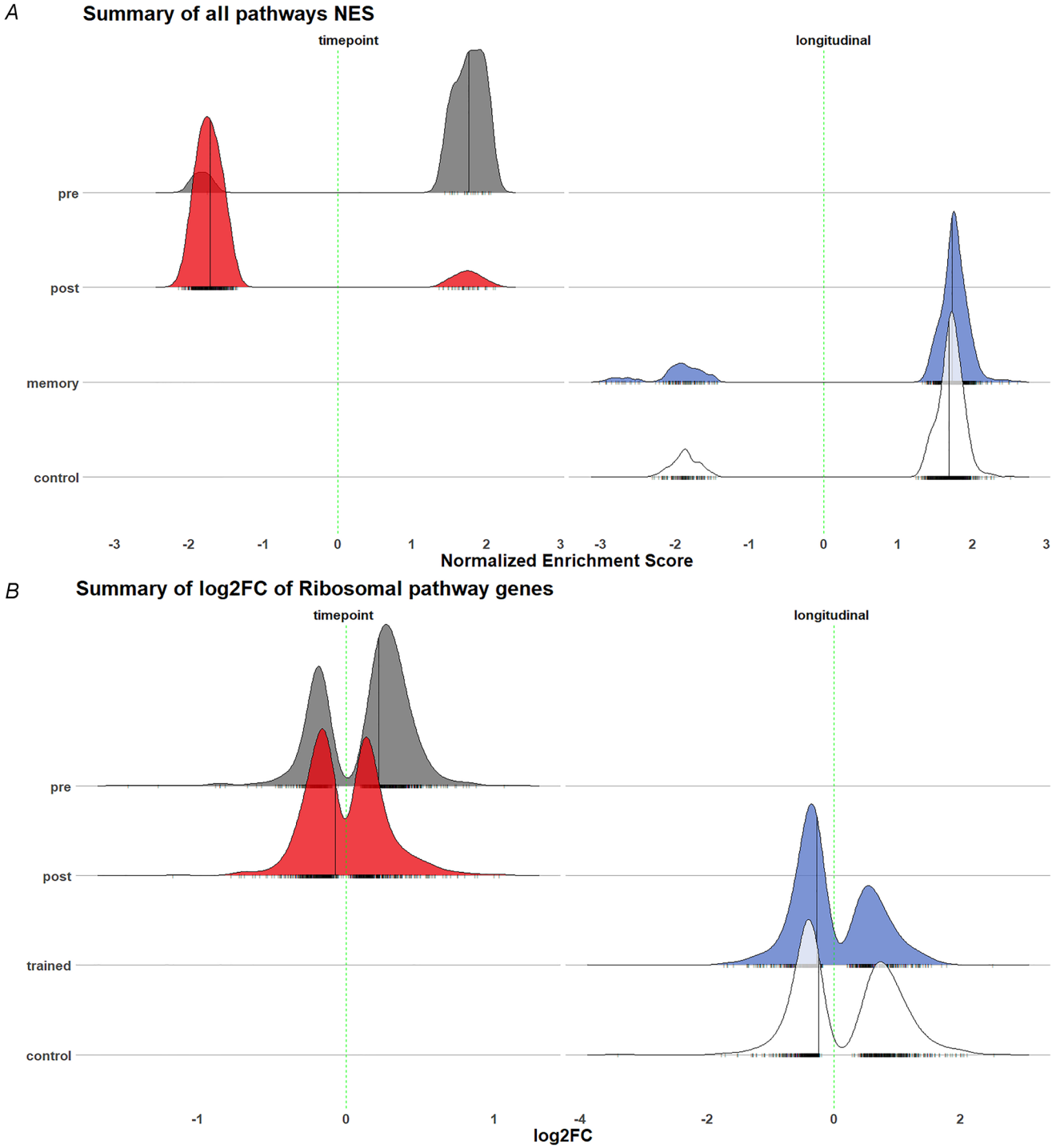

Figure 11. Gene set enrichment results for all pathways
Ridge plot of the summary of gene set enrichment results of all pathways based on normalized enrichment score (NES) (*A*) and pathways associated with the ribosome based on log2 fold change (*B*) Comparisons at the pre (grey) and post (red) time point at the re-training period are based on the previously trained (blue) *vs*. control muscle (white) contrast. Longitudinal comparisons for previously trained and control muscle are based on the post *vs*. pre contrast. The vertical black line in each ridge represents the median.

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strength gain from the first training period was largely retained during de-training. The difference in strength before the re-training period may therefore partly explain why the control arm had a larger gain in strength in the re-training period. Future studies should therefore consider implementing even longer de-training periods to allow for proper reductions in muscle strength.

The lack of faster regain in muscle strength and mass is similar to the observations in some earlier muscle memory studies in humans (Blocquiaux et al., 2020; Psilander et al., 2019; Taaffe & Marcus, 1997). A major difference in those studies compared to our study was, however, that the cellular basis for muscle memory was not established or investigated. Because of the confirmed myonuclear accretion in our study, a faster gain in muscle mass was expected in the previously trained muscle. The previously trained muscle had higher fCSA of the type 2 fibres at the end of the re-training period, which can reflect a faster gain in fibre size. Furthermore, the delta change in fCSA for both type 1 and type 2 fibres was numerically higher in the previously trained *vs*. control muscle, but individual variation, particularly in one of the participants, made the difference between muscles statistically insignificant. In this participant, fCSA in the control muscle increased by ∼50% whereas a decrease was observed in the previous trained muscle (Fig. 6*A*). Although controversial, this participant can be defined as an outlier using Grubbs's test to identify outliers. When this participant was removed from the analyses, a significant difference in favour of the previously trained muscle was observed in fibre types. Furthermore, a closer examination of the individual responses revealed that the myonuclear domains before the re-training period were highly correlated to the individual changes in fCSA during re-training in type 1 fibres and tended to be in type 2 fibres. Consequently, the fibres with the highest density of myonuclei were the ones with the largest degree of hypertrophy during re-training, further supporting the number, or more specifically the density, of myonuclei as an important factor for further muscle growth. In summary, by the initial myonuclear accretion upon the first training period and the maintenance during re-training we were able to investigate if myonuclear accretion may contribute to a beneficial muscle memory for hypertrophy and strength gains in humans. However, because of the large inter-individual variation in the fCSA response and maintained strength during de-training, our data presented here did not enable us to also show significant faster regain of muscle size and strength in the previously trained muscle. Nevertheless, the observed correlation between myonuclear density and the corresponding increase in fCSA in the re-training period supports a possible positive role of the previously accreted myonuclei, which needs to be further studied in humans.

Whether or not newly acquired myonuclei are permanent in humans is highly debated. In contrast to our results, there are some studies observing that myonuclear content is reduced during atrophy, while others report that myonuclear content is not lost but maintained during atrophy (reviewed in Gundersen, 2016; Rahmati et al., 2022; Snijders et al., 2020). Our results clearly support that myonuclei are kept during de-training atrophy, at least within the time period investigated in this study. The two human studies showing reduced number of myonuclei after de-training investigated elderly after 12 weeks and 1 year after an initial training period (Blocquiaux et al., 2020; Snijders et al. 2019). Snijders et al. (2019) observed a 19% reduction in number of myonuclei in two fibres in parallel with an ∼14% non-significant reduction in fibre area. Their method of quantification of myonuclear content did not, however, allow for separation between myonuclei and other nuclei within the basal lamina, and this limitation may explain the seemingly contrasting observations to our results. The study by Blocquiaux et al. (2020) had a methodological approach similar to the present study; however, with only six (exercise) and three (control) participants included in these analyses, the observed decrease in type 1 myonuclei, not accompanied by atrophy, should be interpreted with caution. Lending further support for our observation on the permanence of myonuclei in humans is a recent observation by Nielsen et al. (2023) in current and former users of anabolic androgenic steroids. Despite no difference in fCSA, the former users had more myonuclei in type 2 fibres resulting in lower myonuclear domains than controls. These results indicate that myonuclei accrued during former hypertrophy (steroids + training) are not lost during atrophy 4 years after cessation of steroids. However, based on current knowledge, it remains unclear if myonuclei are lost during longer de-training or immobilisation.

The two previous human studies investigating muscle memory by permanent myonuclear accretion failed to increase myonuclear number/density during the first training period (Blocquiaux et al., 2020; Psilander et al., 2019), which logically makes it impossible to test the memory effect by myonuclear accretion. A possible reason for the successful increase in myonuclear number in the first training period and the fibre atrophy during de-training in our study was the choice of muscle group investigated. While the two former studies trained vastus lateralis, we trained biceps brachii. We chose biceps brachii because it is not used much in daily activities and is not involved in weight bearing activities, which is the case for vastus lateralis. Additionally, we recruited subjects that had not participated in any form for training that should result in hypertrophy of their arm flexors. Thus, the subjects had a similar starting point with no memory of past strength training of their arm flexors.

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Although highly debated (Egner et al., 2017; Englund et al., 2021; McCarthy et al., 2011; McCarthy et al., 2017), myonuclear accretion is in our view regarded as being of key importance for muscle fibre hypertrophy because depletion of satellite cells (Egner et al., 2016) or ablation of satellite cell fusion (Goh & Millay, 2017; Goh et al., 2019), which prevents muscle fibres from acquiring new myonuclei, resulted in no or limited hypertrophy in developing muscles. Therefore, the parallel increase in fCSA in some subjects (non-significant at a group level) and myonuclear number during the first training period, keeping the myonuclear domain unchanged, was as expected and so was the training-induced increase in myonuclear number in the control arm in the re-training period. More surprisingly was the further increase in myonuclear number in both fibre types in the previously trained muscle in the re-training period. Notably, the second increase in myonuclei was lower than the first (33% *vs*. 15% in type 2 fibres), whereas the increase in fCSA was similar (34% *vs*. 30% in type 2 fibres). According to the theory of myonuclear domain, the second increase in number of myonuclei in the previously trained muscle should not be necessary as the fibres only regained the size they had after the first training period (Hall & Ralston, 1989). Consequently, after the re-training period both type 1 and type 2 fibres reached the same fCSA as after the first training period, but because of the additional accretion of myonuclei, the myonuclear domain was lower than before training. This indicates that it is not the size of the fibre *per se* that controls the recruitment of satellite cells for myonuclear accretion. The mechanism might be more related to an increase in size from any starting point or that the training stimulus recruits satellite cells irrespective of the absolute size of the fibre. Interestingly, such a mechanism may enable a large increase in myonuclear number by repeated periods of hypertrophy and atrophy with training–de-training–re-training cycles.

In the present study, despite the significant increase in myonuclear number, the proportion of myonuclei expressed as a percentage of all nuclei in the muscle tissue was roughly reduced from 70% to 60% with training. This indicates a larger increase in in the number of nuclei outside the muscle fibres – interstitial nuclei, which can be the result of, for example, increases in satellite cells (Kadi et al., 2004) and migration of circulatory monocytes and/or endothelial progenitor cells (Strömberg et al., 2017). The percentage of myonuclei observed in our study was slightly lower than the 80% previously reported in human muscle (Bengtsen et al., 2021), and may be due to differences between muscles (vastus lateralis *vs*. biceps brachii) or differences in the methods applied (nuclei sorting). Importantly, the observed changes in interstitial nuclei – increasing during training and decreasing during de-training – might have conflicted with previous results reporting changes in myonuclear content. As mentioned, for studies lacking a specific marker for myonuclei, the quantification of nuclei outside *vs*. inside the fibres is challenging and changes in other nuclei may therefore have contributed to the inconsistent results regarding myonuclear permanence after de-training/disuse. This might be the case in studies that report that myonuclei are not maintained during atrophy in mice (Dungan et al., 2019; Murach et al., 2020).

The translation of the observed myonuclear accretion and permanence to the macroscopic level is unclear from the present data. For this reason, the hypothesis of muscle memory by nuclear accretion should be further investigated in parallel with alternative mechanisms such as motor learning and epigenetics. Since the present study has a low number of participants, we encourage future studies to include more than in the present study due to the large individual variations seen in untrained participants.

Transcriptional profile

The present study also aimed at investigating molecular factors that might aid the trained muscle upon re-training. Overall, the previously trained and (untrained) control muscles showed differences in gene expression following a period of de-training, but the differences were mostly evened out after the re-training period. In training-naïve muscle, the effects of resistance training tend to be larger and more related to stress responses than in highly experienced skeletal muscle (Damas et al., 2018). Accordingly, the control muscle was expected to show a larger number of differentially regulated genes in the re-training period than the previously trained muscle. The previously trained muscle, having undergone this initial large training effect in the first training period would, besides a number of 'core genes', show a lower number of genes differentially regulated while the control muscle is catching up in the re-training period (Fig. 9). The trained muscle being ahead of the control muscle is also reflected on the pathway level, where both muscles generally increased the normalized enrichment (NES) of pathways after the re-training period compared to before. Notably, comparing muscles at these time points, the enrichment balance lies on the side of the previously trained muscle before, but on the side of the control muscle after re-training (Fig. 11*A*), indicating the stimulation of a larger number of pathways during an initial training period.

Importantly, at the start of the re-training period, the previously trained muscle showed a significant enrichment in pathway terms related to the ribosome and the protein production machinery (Fig. 10*A*). Increased translational capacity has previously been linked to the hypertrophic potential in skeletal muscle (Millward et al., 1973; Stec et al., 2016), also suggested to contribute to

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a potential muscle memory mechanism (Moberg et al., 2020). When comparing by pathway level analysis, an increased translational potential could be reflected by the median fold change of ribosome-pathway related genes being on the side of the previously trained muscle before the re-training period (Fig. 11*B*). Such an adaptation, which probably was a consequence of the training protocol employed in the first training period, seems to be long lasting because it was present even following the 16-week de-training period. This can be translated into an increased potential for protein synthesis during the re-training period and, in theory, potentially contribute to favourable adaptations in the previously trained muscle leading to faster hypertrophy. As already discussed, this was not the case in our study, but a faster hypertrophy in the previously trained muscle might have been masked by the extreme response noted in one outlier. As expected, the control muscle is catching up on the ribosome-pathway-related genes after the re-training period as the median log2FC expression comparing both conditions shifts close to zero and with that towards similar expression as in the previously trained muscle which is only recently confronted with a training stimulus that can increase the translational potential (Fig. 11*B*).

In our study, we further identified three genes that were differentially expressed comparing previously trained *vs*. control muscles following a de-training period and could therefore be involved in a molecular muscle memory. The transcription factor early growth response 1 (*EGR1*) has previously been reported to influence a large range of cellular proliferation and differentiation processes (Zhang et al., 2018) and is therefore an interesting target for the orchestration of a molecular memory mechanism. Further, *EGR1* is an important transcription factor for memory formation in the brain and acts together with ten–eleven translocation (Tet) proteins which cause DNA demethylation in promotor regions allowing the transcription of the downstream genes (Sun et al., 2019). A more hypomethylated DNA signature has already been established as one possible memory mechanism after strength training (Seaborne et al., 2018; Wen et al., 2021b), and *EGR1* can therefore potentially play a central role in epigenetic memory formations in muscle. Notably, *EGR1* expression was lower in the previously trained muscle than in the control muscle after the de-training period. This seemingly counterintuitive finding may be explained by an earlier upregulation of *EGR1* during the first training period when changes in DNA methylation probably do occur. The downregulation observed after the de-training period is consistent with observations of reduced *EGR1* expression levels in tendons with reduced loading and indicates that *EGR1* expression is mechanosensitive (Gaut et al., 2016). An earlier upregulation of *EGR1* is supported by the finding of higher expression levels of *COL1A1* in the trained muscle than in the untrained muscle because *EGR1* has been shown to bind to the promotor region of *COL1A1* and activate gene expression, for example, during tendon differentiation (Guerquin et al., 2013). The long-term increase in *COL1A1* expression might be explained by maintained hypomethylation in the promotor region during de-training (Blocquiaux et al., 2022). The significant increase in *EGR1* expression level in the previously trained muscle during the re-training period supports a general stimulation of EGR1 by strength training. This also correspond to the known stimuli which induce *EGR1* expression like mechanical stress, hypoxia, growth factors, hormones and cytokines and seems to induce *EGR1* expression also in other exercise-relevant tissue such as tendons, cartilage, bone and vessels (Havis & Duprez, 2020). Keeping this in mind, we cannot rule out that the differences in gene expression between the two muscles were related to changes in the content of interstitial nuclei in the previously trained muscles. In mice, it has been shown that the epigenetic changes during training and de-training are different between myonuclei and interstitial nuclei (Wen et al., 2021a). Consequently, future studies in humans should strive to differentiate between myonuclei and non-myonuclei when investigating differences in gene expression and epigenetic modifications in muscle memory protocols.

However, there might be a direct link between the regulation of EGR1 and the observed increases in myonuclear number because EGR1 has recently been identified as regulator of myogenin, promoting satellite cell differentiation in a series of overexpression and knock-down experiments in bovine skeletal muscle (Zhang et al., 2018). As already discussed, we believe that there has been a transient early upregulation of EGR1 with strength training in the first training period. Importantly, although EGR1 was downregulated with de-training the suggested hypomethylation caused by an early and transient EGR1 activation may be long lasting and serve as the epigenetic memory tag observed by others (Seaborne et al., 2018; Wen et al., 2021b). Together with our results on the increased number of myonuclei, this could indicate a role for EGR1 in myonuclear accretion by regulating the differentiation and fusion of satellite cells into the growing muscle fibres during strength training. Interestingly, further exercise during re-training increased the myonuclear accretion, and in this period the increased myonuclear number was paralleled by increased *EGR1* expression in the previously trained muscle, but not in the control muscle. These results might indicate that *EGR1* expression levels are rapidly and transiently increased by exercise in untrained muscles, whereas expression levels are initiated at a later stage or are more long lasting in previously trained muscles.

The *COL1A1* gene encodes the collagen type I alpha 1 chain – one of the chains in collagen type I; it and collagen III are the most abundant collagen types in fibrous part of

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the extracellular matrix (ECM) (Csapo et al., 2020). The increased expression of *COL1A1* in the trained muscle before the re-training period, and after the re-training period in both muscles, corresponds well with the suggested transient training-induced increases in *EGR1* expression levels inducing a demethylation in the *COL1A1* promoter region. In fact, Blocquiaux et al. (2022) recently reported enhanced *COL1A1* expression in muscles with a previous history of training, and this observation corresponded perfectly with reduced methylation in the promotor region. The increased expression of *COL1A1* is interesting for its role in focal adhesion and structural integrity of the ECM, which could improve activation of the mechano-sensing pathway, improving progress in a retraining context (Blocquiaux et al., 2020). It is plausible that changes in gene expression do not originate from muscle cells, but from other important cells in skeletal muscle such as the fibroblasts, which are the main producer of proteins making up the ECM (Chapman et al., 2016). Interestingly, fibro-adipogenic progenitors (FAPs) are activated and proliferate in response to overload induced hypertrophy and seem to play a role in both satellite cell activation and strengthening of the ECM (Brightwell et al., 2022). One subpopulation of FAPs identified in human muscle is characterised by high expression levels of *COL1A1* and *COL1A2*, and a possible increase in this population may have contributed to the general increase in interstitial myonuclei and the long lasting increased expression levels of *COL1A1* in our study (Negroni et al., 2022). Further, a structural strengthening of the ECM can be long lasting and theoretically also contribute to increased strength during training and maintained muscle strength in the de-training period (Csapo et al., 2020). The physiological and functional role of the increased *COL1A1* expression and the possible role of FAPs in this context is, however, speculative and needs to be further investigated.

The gene for myosin light chain 5 (*MYL5*), part of the actin–myosin contraction complex, was the third gene significantly increased in the trained muscle compared to the control muscle after the de-training period. In contrast to our finding, other MYL genes (*MYL1*, *MYL2*, *MYL3*) are downregulated by inactivity (Pillon et al., 2020) and it is not clear how the long-lasting increased expression of *MYL5* might impact muscle function. It has been shown that the expression of the different *MYL* genes is fibre type dependent and the MYL5 protein is expressed in adult type 1 fibres, although at low levels (Murgia et al., 2021). Notably, the difference in expression seen in our study was not related to changes or differences in fibre type composition (39% type 1 fibres in both muscles) because fibre distribution was not different between muscles, nor did it change in response to training (data not shown). Two alternative and highly interesting roles of *MYL5* are related to general cell division and bidirectional gene regulation with HIF-1α (Ramirez et al., 2021; Zhang et al., 2017). Although speculative, these observations indicate a possible role of MYL5 in satellite cell proliferation potentially contributing to the observed increase in myonuclear number. A causal relation was not investigated in our study, but the increased expression of *MYL5* with training in both muscles was paralleled with an increase in myonuclear number in the re-training period. A possible role of *MYL5* related to hypertrophy, the memory of hypertrophy and formation of new myonuclei is highly interesting, but a possible causal relation needs further investigation.

Conclusion

Our results showed that myonuclear number was increased in the first training period and maintained during 16 weeks of de-training. This finding forms the basis of investigation into whether myonuclear accretion of previous training increased the potential for enhanced hypertrophy, e.g. muscle memory. Additionally, three genes that were long-term differentially regulated in the previously trained compared to the control muscle after de-training may indicate an additional transcription-regulatory memory mechanism, but only descriptive changes were available in this study. Importantly, the increased number of myonuclei and the differentially regulated genes in the previously trained muscle did not translate into significant superior muscle growth or faster increase in muscle strength compared to the control muscle in the re-training period. Thus, the physiological memory response in human muscle remains to be determined by future studies.

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Additional information

Data availability statement

The data that support the findings of the present study are available from the corresponding author upon reasonable request.

Competing interests

The authors declare no conflicts of interest in relation to the data presented in this manuscript.

Author contributions

K.T.C., O.R.S., N.P., T.R. conceived and designed the study. K.T.C., M.H., K.S. performed the experiments. K.T.C., S.M.R., M.H., K.S., O.H. analysed data. K.T.C., S.M.R., M.H., K.S., O.R.S., O.H., N.P., C.J.S., T.R. interpreted the results. K.T.C., S.M.R., T.R. created the figures and drafted the manuscript. All authors edited, revised, and approved the final version of the manuscript submitted for publication, and agree to be accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved. All who designated as authors qualify for authorship, and all who qualify for authorship are listed.

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Keywords

CSA, gene expression, muscle fibre, skeletal muscle

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